

Performance Testing

Date	02 October 2026
Team ID	6ab7f6f8c23107aaaa72871e
Project Name	FitBuddy - AI Fitness Plan Generator using Gemini Models
Maximum Marks	5 Marks

Performance Testing

Performance testing for the submission focuses on the documented local execution path. The source materials do not contain formal load-test measurements, so no unverified response-time or concurrency claims are added.

Step 1: Testing Overview

Field	Details	Acceptance / Note
System	FastAPI + Uvicorn local server	
Test environment	Local development environment with Python virtual environment and required packages	
API dependency	Google Gemini API configured through GOOGLE_API_KEY	
Primary pages	/, result flow, /docs, /view-all-users	
Testing status	Functional flow verified from the documented implementation path; formal load benchmark not recorded.	

Step 2: Test Scenarios

ID	Scenario	Input / Action	Expected Result	Status
T-01	Profile submission	Enter valid name, user ID, age, weight, goal and intensity.	Plan generation route receives expected fields.	Pass
T-02	Workout generation	Trigger /generate-workout.	Structured 7-day plan is returned to result.html.	Pass
T-03	Nutrition tip	Generate a plan for a selected goal.	Concise goal-aligned tip is shown.	Pass
T-04	Feedback update	Submit a feedback message with user ID.	Updated plan is generated and stored.	Pass

ID	Scenario	Input / Action	Expected Result	Status
T-05	Admin view	Open /view-all-users.	Stored user and plan records are listed.	Pass
T-06	API docs	Open /docs.	Interactive FastAPI docs load locally.	Pass
T-07	Invalid / missing AI key	Run without GOOGLE_API_KEY.	Configuration error should be identified during setup rather than silently failing.	Configuration check

Performance note

Formal throughput, latency and concurrent-user measurements are not documented in the source materials. Before production-scale deployment, load testing and capacity benchmarking should be added.